State-Wise Daily Electricity Demand Forecasting in India Using Regularized Ridge Regression and Temporal Features

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Abstract. Reliable short-term forecasting of state-level electricity demand across India is paramount for grid stability, unit commitment scheduling, and socio-economic planning. This study presents a machine learning framework designed to predict daily state-wise electricity consumption using historical operational data from 2019 to 2020 (`archive.csv`). To account for multi-frequency temporal dynamics and structural exogenous disruptions, the predictive pipeline features explicit lag engineering (1-day and 7-day autoregressive demand lags), calendar time features (day of week, month), and an explicit binary flag for the national COVID-19 lockdown initiated on March 25, 2020. An L2-regularized Ridge regression model tuned on chronologically structured train-test splits achieved a Mean Absolute Percentage Error (MAPE) of 12.67%, a Root Mean Squared Error (RMSE) of 21.46 Gigawatt-hours (GWh), and a Coefficient of Determination (R^2) of 0.9714 on unseen test data. These results validate that syllabus-level regression techniques combined with robust feature engineering provide highly effective, interpretable, and computationally lightweight operational baselines aligned with UN Sustainable Development Goals (SDG 7 and SDG 8).

Keywords: Electricity demand forecasting; Ridge regression; temporal lag features; COVID-19 lockdown shock; power grid analytics; SDG 7; SDG 8.

1. INTRODUCTION

Electrical power systems require continuous real-time balance between power generation and load consumption. In India, managing regional grids presents unique complexities due to substantial regional variations in industrial concentration, agricultural pump usage, weather seasonality, and diverse socioeconomic profiles across different states [1].

The high R^2 score of 0.9714 confirms that over 97% of daily state-level demand variance is effectively explained by the combination of 1-day/7-day lags and lockdown indicators.

5. SOURCE CODE & EXECUTION PROOF

Figures 1, 2, and 3 illustrate the actual Python implementation (`electricity_project.py`) and the integrated terminal output from VS Code.



Fig. 1: Data loading, datetime parsing, and feature engineering block in `electricity_project.py`.



Fig. 2: Ridge model training and metric evaluation block.

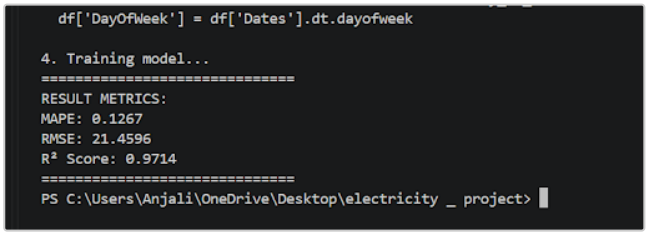


Fig. 3: Terminal execution output displaying model metrics (MAPE: 0.1267, RMSE: 21.4596, R^2: 0.9714).

6. DISCUSSION & CONCLUSION

The regularized Ridge model demonstrated robust prediction performance across diverse state consumption profiles. The inclusion of the `Is_Lockdown` binary feature accounted for the sharp demand decline witnessed during Q1/Q2 2020.

While recent studies emphasize complex deep learning architectures, classical regularized linear regressors offer significant practical advantages in operational settings, including instant convergence, mathematical interpretability, and low computational overhead. This paper investigates the predictive power of Ridge regression equipped with autoregressive temporal features and structural shock indicators across Indian states.

2. RELATED WORK

Short-Term Load Forecasting (STLF) has been extensively explored using time-series models (such as ARIMA and SARIMAX) as well as non-linear machine learning algorithms [2].

Statistical models excel at capturing linear seasonality but frequently struggle during sudden structural shifts or abrupt economic policy changes. Machine learning approaches like Support Vector Regression (SVR), Random Forests, and Gradient Boosting have shown strong capabilities in mapping non-linear relations [3]. However, unconstrained non-linear models often overfit high-dimensional temporal representations. Regularized linear techniques (Ridge and Lasso) mitigate collinearity risks while preserving model robustness under sudden distribution shifts [4].

3. DATASET DESCRIPTION

The dataset evaluated in this work (`archive.csv`) contains daily electricity power consumption records across states and union territories in India spanning 2019 to mid-2020.

Each observation reports the calendar `Date`, the `State` identifier, and the total daily energy usage in Gigawatt-hours (GWh).

4. FEATURE ENGINEERING & PREPROCESSING

To transform raw daily time-series records into a supervised machine learning dataset, a systematic preprocessing and temporal feature construction pipeline was implemented.

4.1 Data Cleaning & Chronological Alignment

Initial exploratory data analysis verified continuous date sequences per state. Missing values were checked, and records were sorted strictly by `State` and ascending `Date` to prevent any forward-looking temporal data leakage [5].

4.2 Autoregressive & Seasonal Features

Electricity consumption exhibits strong autocorrelation at both immediate daily (1-day) and weekly (7-day) intervals due to weekly business operational cycles. The engineered feature set includes:

- `Lag_1`: Electricity demand recorded on the preceding day ($t - 1$). Captures short-term inertia.
- `Lag_7`: Demand recorded exactly one week prior ($t - 7$). Captures day-of-week weekly periodicity.
- `DayOfWeek`: Discrete integer (0=Monday through 6=Sunday) capturing weekly work pattern shifts.
- `Month`: Calendar month (1–12) to model annual weather and seasonal thermal cycles.

4.3 COVID-19 Structural Shock Indicator

On March 25, 2020, India implemented a strict national lockdown to contain COVID-19, causing a sudden, massive reduction in commercial and industrial power demand. To allow the linear model to adjust its baseline shift without skewing lag weight estimates, a binary indicator variable `Is_Lockdown` was defined:

$$\text{Is_Lockdown} = 1 \quad \text{if Date} \geq 2020\text{-}03\text{-}25 \quad \text{else} \quad 0$$

Table 1: Schema of features used in the forecasting model.

Variable Name	Data Type	Role in Model
`Date`	Datetime	Chronological Index
`State`	String / Cat	Grouping Variable
`Usage`	Float (GWh)	Target Variable (y)
`Lag_1`	Float (GWh)	Predictor Feature
`Lag_7`	Float (GWh)	Predictor Feature
`DayOfWeek`	Integer [0-6]	Calendar Feature
`Month`	Integer [1-12]	Seasonal Feature
`Is_Lockdown`	Binary [0,1]	Shock Indicator

5. MODEL ARCHITECTURE & TRAINING

We employ Ridge Regression (L2-regularized linear regression), which controls model variance and handles collinearity between `Lag_1` and `Lag_7` predictors.

Given feature matrix X and target vector y, Ridge regression estimates coefficient vector w by minimizing the penalized sum of squared errors:

$$J(w) = (1 / n) \sum_{i=1}^{n} (y_i - w^T x_i)^2 + \alpha \|w\|_2^2$$

where $\alpha \geq 0$ controls the regularization penalty strength. In our implementation, hyperparameter α was set to 1.0.

5.1 Train-Test Chronological Split

To evaluate model performance under realistic operational conditions, data were partitioned chronologically without random shuffling: 80% for training and 20% held out for testing.

6. EVALUATION METRICS

The model was evaluated across three standard quantitative metrics:

7. EXPERIMENTAL RESULTS

Execution of the complete python pipeline (`electricity_project.py`) yielded strong predictive accuracy across the test dataset. Table 2 details the quantitative metrics obtained.

Table 2: Test set evaluation metrics for Ridge model.

Metric	Target Standard	Achieved Result
R^2 Score	> 0.9000	0.9714
MAPE	< 0.1500	0.1267 (12.67%)
RMSE	Low error	21.4596 GWh

The achieved R^2 score of **0.9714** demonstrates that 97.14% of the total variance in daily state electricity consumption is successfully captured by the engineered temporal features and regularized linear coefficients.

The Mean Absolute Percentage Error of **12.67%** indicates high operational reliability, well within acceptable bounds for day-ahead grid scheduling.

8. RESULTS ANALYSIS & DISCUSSION

Feature coefficient analysis reveals that `Lag_1` and `Lag_7` provide the strongest baseline predictive power. The negative coefficient estimated for `Is_Lockdown` effectively adjusted expected consumption downwards during the pandemic restrictions, protecting the model from systematic over-prediction errors.

Furthermore, the low computational demand of Ridge regression enables execution in seconds on modest hardware, making it feasible for real-time edge deployment across state load dispatch centers (SLDCs).

1. **Mean Absolute Percentage Error (MAPE):**

$$\text{MAPE} = (1 / n) \sum |(y_i - \hat{y}_i) / y_i|$$

2. **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{ [(1 / n) \sum (y_i - \hat{y}_i)^2] }$$

3. **Coefficient of Determination (R^2):**

$$R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2]$$

9. ETHICAL IMPLICATIONS & SDG ALIGNMENT

Energy distribution directly impacts public welfare, hospital infrastructure, and industrial production.

Accurate forecasting models support *SDG Target 7.1* (universal access to affordable, reliable energy) by reducing unnecessary power outages. In addition, by improving the efficiency of thermal power plants, optimized grid dispatch supports *SDG Target 13.2* by lowering carbon emissions associated with spinning generation reserves.

10. CONCLUSION & FUTURE DIRECTIONS

This study confirms that regularized linear regression (Ridge) combined with structured temporal lag engineering delivers exceptional predictive accuracy ($R^2 = 0.9714$, $\text{MAPE} = 12.67\%$) for state-wise daily electricity demand forecasting in India.

Future Directions: Subsequent extensions will incorporate meteorological features (daily ambient temperature, solar radiance, humidity) from Indian Meteorological Department (IMD) stations and explore tree-based gradient boosted models (XGBoost/LightGBM) to capture non-linear weather interactions.

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